

```
PLAY [Configure bootstrap Consul node] *****

TASK [Gathering Facts] *****
ok: [host1]

TASK [Create Consul bootstrap directory] *****
changed: [host1]

TASK [Copy configuration file] *****
changed: [host1]

TASK [Copy systemd bootstrap consul.service] *****
changed: [host1]

TASK [Start bootstrap Consul] *****
changed: [host1]

PLAY [Configure other Consul nodes] *****

TASK [Gathering Facts] *****
ok: [host2]
ok: [host3]

PLAY RECAP *****
host1    : ok=6    changed=4    unreachable=0    failed=0
host2    : ok=2    changed=0    unreachable=0    failed=0
host3    : ok=2    changed=0    unreachable=0    failed=0
```

Other Consul nodes

The last step is to configure the other Consul nodes in the cluster. A `/etc/consul.d/server` directory is created and the node configuration file is copied to it. The contents of the configuration file are as follows:

```
{
  "bootstrap": false,
  "server": true,
  "datacenter": "chennai",
  "data_dir": "/var/consul",
  "encrypt": "PfCkrq/lFPem7slKP5N2Cw==",
  "log_level": "INFO",
  "enable_syslog": true,
  "start_join": [ {% for item in groups['all'] %} "{{
hostvars[item].ansible_default_ipv4.address }}" {% if not
loop.last %} , {% endif %} {% endfor %} ]
}
```

The configuration specifies that this is not a bootstrap node but is part of the Consul cluster. The data centre, data directory and encryption keys are specified. The log level information is also supplied. Finally, the IP addresses of all the Consul nodes are mentioned to join the cluster.

A `systemd` configuration file is also created to start the Consul service on these nodes, as shown below:

```
[Unit]
Description=Consul service discovery agent
Requires=network-online.target
After=network-online.target

[Service]
Environment=GOMAXPROCS=2
Restart=on-failure
ExecStart=/usr/local/bin/consul agent -server -data-dir=/
tmp/consul -node={{ inventory_hostname }} -config-dir=/etc/
consul.d/server -enable-script-checks=true
ExecReload=/bin/kill -HUP $MAINPID
KillSignal=SIGTERM

[Install]
WantedBy=multi-user.target
```

The `inventory_hostname` is used as the node name when starting the Consul service. The playbook for setting up the Consul nodes is given below:

```
- name: Configure other Consul nodes
  hosts: server
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [nodes]

tasks:
  - name: Create Consul server directory
    file:
      path: /etc/consul.d/server
      state: directory
      mode: 0755

  - name: Copy configuration file
    template:
      src: "../../files/node-config.json.j2"
      dest: /etc/consul.d/server/config.json

  - name: Copy systemd node consul.service
    template:
      src: "../../files/node-consul.service.j2"
      dest: /etc/systemd/system/consul.service

  - name: Start Consul service
    systemd:
      name: consul.service
      state: started
```

The above playbook can be invoked as follows:

```
$ ansible-playbook -i inventory/kvm/inventory playbooks/
configuration/consul.yml --tags nodes -vv -K
```